# BHOOPENDRA KUMAR

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## ACADEMIC PROFILE

Assistant Professor in the Department of Computer Science and Engineering at Vivekanand Institute of Professional Studies, Delhi, since August 2026. Previously served as Assistant Professor at SRM University Haryana and PSIT Kanpur, with earlier industry experience as an Associate Analyst. Academic and research interests include information security, financial fraud detection, privacy-preserving computation, and computer networks.

## RESEARCH INTERESTS

Information Security, Financial Fraud Detection, Privacy-Preserving Computation, Computer Networks.

## ACADEMIC AND PROFESSIONAL APPOINTMENTS

### Assistant Professor | Vivekanand Institute of Professional Studies, Pitampura, Delhi | August 2026–Present

- Department of Computer Science and Engineering; current academic appointment.
- Courses taught: Python Programming and Machine Learning.

### Assistant Professor | SRM University Haryana | September 2025–30 July 2026

- Worked in the Computer Science Department until 30 July 2026.
- Courses taught: Data Structures and Computer Networks.

### Assistant Professor | PSIT Kanpur | July 2024–September 2025

- Worked in the Department of Computer Science and Engineering (CSE).
- Courses taught: Python Programming and Data Structures.

### Associate Analyst | GlobalLogic, Gurugram | July 2021–July 2022

- Worked on the Google Merlot project.
- Analyzed e-commerce website data as part of the Google Merlot project to support project-related data review and analysis.

## EDUCATION

M.Tech, Computer Science and Information Security | Malaviya National Institute of Technology Jaipur | 2024 | CGPA: 7.17

B.Tech, Computer Science and Engineering | Veer Bahadur Singh Purvanchal University, Jaunpur | 2019 | 62.20%

Intermediate | Jawahar Navodaya Vidyalaya, Unnao | 2014 | 76.20%

Matriculation | Jawahar Navodaya Vidyalaya, Unnao | 2012 | CGPA: 9

## RESEARCH PUBLICATIONS

### Journal Manuscript

1. “Noise Analysis in Quantum Neural Network.” Quantum Information Processing. Status: Under Review (as listed in source CV).

- Research problem: Investigates the effect of hardware noise, particularly depolarizing gate noise, on the reliability, training convergence and classification performance of quantum neural networks in the Noisy Intermediate-Scale Quantum (NISQ) era.
- Proposed methodology: Developed a hybrid quantum–classical neural network for binary and multi-class classification. Principal Component Analysis (PCA) reduces classical input features to quantum-compatible dimensions; angle encoding and entangling CNOT layers form the variational quantum circuit; quantum measurements are coupled with a classical fully connected prediction layer.
- Experimental design: Implemented differentiable quantum simulation in PennyLane, injecting depolarizing noise into encoding and variational gates. Evaluated classification and training behavior on the Iris and Wine datasets across noise probabilities from 0.00 to 0.20.
- Key findings and contribution: Observed stable convergence and strong classification performance under low-to-moderate noise, followed by gradual deterioration of test accuracy and training loss as noise increased. The study offers evidence and practical guidance for noise-aware design of shallow, PCA-reduced hybrid QNNs for near-term hardware.

### Conference Publications and Accepted Papers

2. “VAE-TF: A Variational Autoencoder Framework for Financial Fraud Detection.” 2025 International Conference on Electronics, AI and Computing (EAIC). Published. DOI: 10.1109/EAIC66483.2025.11101301. ● Research focus: Financial fraud detection using a variational autoencoder framework.

3. “Post-Quantum Secure MPC Framework for Confidential Neural Network Inference.” Status: Accepted (as listed in source CV).

- Research focus: Confidential neural network inference using post-quantum security and secure multiparty computation (MPC).

## M.TECH THESIS / RESEARCH PROJECT

### Image Encryption Scheme Using Chaotic Maps | 2024

● Developed an image encryption scheme using confusion and diffusion operations applied to an image pixel matrix. ● Implemented image matrix processing using Python libraries.

## TEACHING EXPERIENCE AND ACADEMIC CONTRIBUTIONS

- Teaching appointments: Vivekanand Institute of Professional Studies (August 2026–Present); SRM University Haryana (September 2025–30 July 2026); PSIT Kanpur (July 2024–September 2025).
- Courses taught across appointments: Python Programming, Data Structures, Computer Networks, and Machine Learning.

## TECHNICAL SKILLS

Programming: C/C++, Python | Web Development: HTML, CSS

**Core Subjects:** Operating Systems, Computer Networks, Data Structures, Database Management Systems